

AI-Powered Smart Farming & Gardening Assistant

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Abstract

The AI-Powered Smart Farming & Gardening Assistant is an intelligent web-based platform designed to support both farmers and home gardeners by leveraging Artificial Intelligence (AI), Machine Learning (ML), IoT, weather intelligence, and data analytics. The system provides personalized recommendations, predictive insights, and real-time monitoring to help users improve plant health, optimize resource usage, increase productivity, and make informed decisions throughout the cultivation process.

The platform is divided into two primary modules: Smart Farming and Smart Gardening. The Smart Farming module focuses on crop recommendation, yield and profit prediction, weather-aware irrigation scheduling, soil health analysis, disease and pest detection, market intelligence, farm management, and predictive analytics. It also integrates IoT devices such as soil moisture, temperature, humidity, and light sensors to enable real-time field monitoring and automated irrigation.

The Smart Gardening module is designed for home gardeners and plant enthusiasts. It offers AI-powered plant identification, plant health assessment, personalized care recommendations, watering and fertilization schedules, disease detection, growth tracking, smart pot monitoring, and gardening planners. Users can also discover nearby nurseries, compare plant prices, access a comprehensive plant encyclopedia, and interact with an AI gardening assistant for instant guidance.

The application combines computer vision to identify plants and detect diseases from uploaded images, machine learning models to predict crop yield, profitability, and risks, and weather APIs to provide climate-aware recommendations. Interactive dashboards visualize plant health, sensor readings, growth trends, irrigation schedules, and farm performance.

To further enhance user experience, the platform includes multilingual support, voice-based interaction, push notifications, report generation, community forums, and an integrated marketplace that connects growers with buyers and gardening resources.

Overall, the AI-Powered Smart Farming & Gardening Assistant aims to promote sustainable agriculture and smart gardening practices by combining modern AI technologies with practical farming and gardening knowledge. By enabling early disease detection, optimizing irrigation, improving soil management, predicting yields and market trends, and delivering personalized care recommendations, the system helps users reduce costs, conserve resources, and achieve healthier plants and higher productivity.

Technologies

Frontend (Mobile + Web): React Native with Expo, react-native-web for web support, NativeWind for styling, React Navigation for screens, Zustand for state management, React Query for API calls, Victory Native or Recharts for charts, React Hook Form for forms, Expo Camera for photo uploads, OpenStreetMap for maps, Expo Notifications for alerts.

Backend: FastAPI (Python), since it connects easily with ML models. Uses JWT for authentication, WebSockets or MQTT for real-time sensor updates, and Celery with Redis for background tasks.

Database: PostgreSQL, hosted free on Supabase. Redis for caching. TimescaleDB extension for storing sensor readings over time.

Machine Learning and Deep Learning: Scikit-learn and XGBoost for crop, yield, and profit prediction. TensorFlow or PyTorch for plant disease detection and plant identification. OpenCV and Deep learning for image processing. Google Colab used for free GPU training. TensorFlow Lite for on-device mobile inference.

IoT: ESP32 microcontroller with soil moisture, temperature, humidity, light, and rain sensors. MQTT (Mosquitto broker) used for real-time communication with the server. Arduino IDE or PlatformIO for programming the ESP32.

External APIs: OpenWeatherMap or Open-Meteo for weather data, OpenStreetMap and Nominatim for location and nursery search, Trefle or Perennial for plant reference data.

Hosting and Deployment: Render or Railway for the backend, Vercel for the web build, Supabase for database and file storage, Expo EAS for building and distributing the mobile app.

Dev Tools: GitHub for version control, Postman for API testing, Figma for UI design, Excalidraw or draw.io for diagrams, Notion for documentation.